A Taxonomy for Analyzing Dashboard Design in XR based Content and Data Visualization Tool

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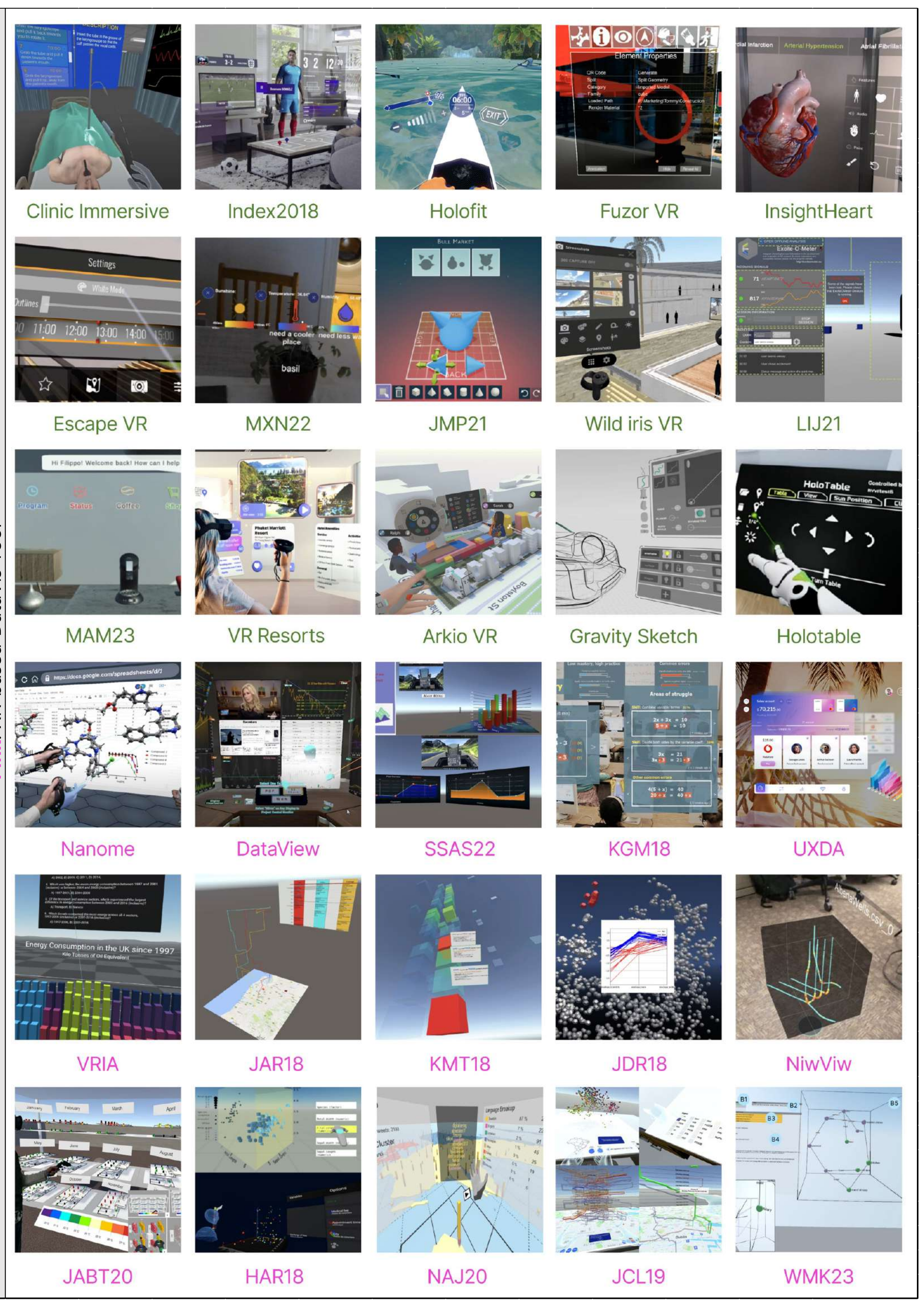
Main category	Sub category level 1	Sub category level 2	Clinic Immersives	Index2018	Holofit	Fuzor VR	Insight Heart	Enscape VR	MXN22	JMP21	Wild Iris VR	LJJ21	MAM23	VR Resorts	Arkio VR	Gravity Sketch	Holotable	Nanome	DataView	SSAS22	KGM18	UXDA	VRIA	JAR18	KMT18	JDR18	NiwViw	JABT20	HAR18	NAJ20	JCL19	WMK23	XR Content (COUNT)	XR-based DataVis Tool (COUNT)
Type of Data (Covered Field)	Categories of Data	Business																															2	1
		Finance																															9	6
		Architecture																															7	7
		Educational																															11	11
		Medical																															2	1
Audience	Types of Audiences in terms of Utilization (group)	Public																															2	5
		Organization																															0	1
		Individual																															8	7
Dashboard Implement	Intermediation in Dashboards	VR																															6	12
		AR																															6	2
		MR																															3	2
	Interaction Device	Joystick																															5	4
		Controller																															4	8
		Hand																															6	4
Purpose of Dashboard	Strategic Dashboard	Showing Directional																															3	4
		Representing Temporal Data																															3	7
		Evaluating																															11	5
		Monitoring																															12	11
	Analytical Dashboard	Providing Context																															3	7
		Determine Cause and Effect																															3	9
	Operational Dashboard	Verify Result and Quick Response																															7	2
		View Real-Time Event																															8	5
		Providing Details																															6	5
Dashboard Design	Layout	Hierarchical																															1	4
		Categorical																															7	6
		Grouped																															12	7
		Schematic																															0	2
	Graphic and Metric	Graph																															8	10
		Image																															8	5
		Icon and Pictogram																															12	7
		Text																															12	11
		Direction Controllers																															0	1
		Table/List																															5	7
	Color	Distinguishing Data Encodings																															9	11
		Differentiating Colors for Different Users																															2	0
		Theme																															4	3
	Dynamic Media	Video																															2	1
		Flash																															2	3
	Behavioral Design	Gesture-driven Dashboard Output/Movement																															1	5
Spatial Design	Dashboard Layer	Single Layer																															12	5
		Multi-layer																															3	10
	Dashboard Placement	Target Centered Lock																															11	12
		Controller Centered Lock																															3	3

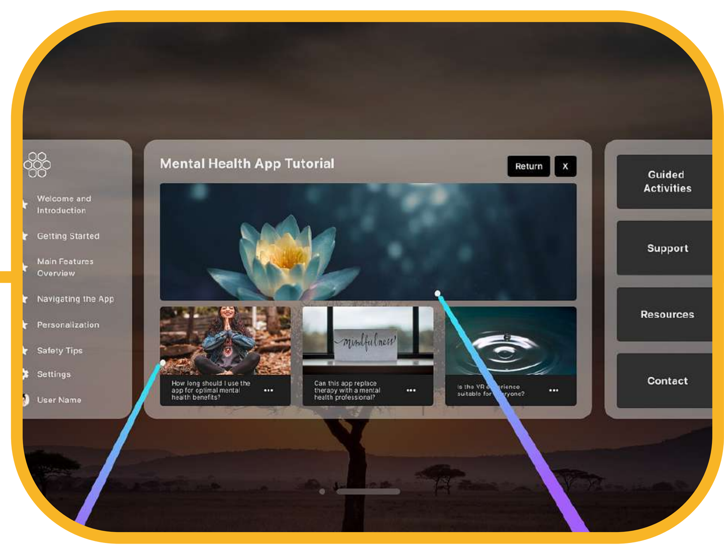
Green: XR Content
Pink: XR-based DataVis Tool



A taxonomy for analyzing dashboard design in XR based content and data visualization tool and representative image of 30 cases

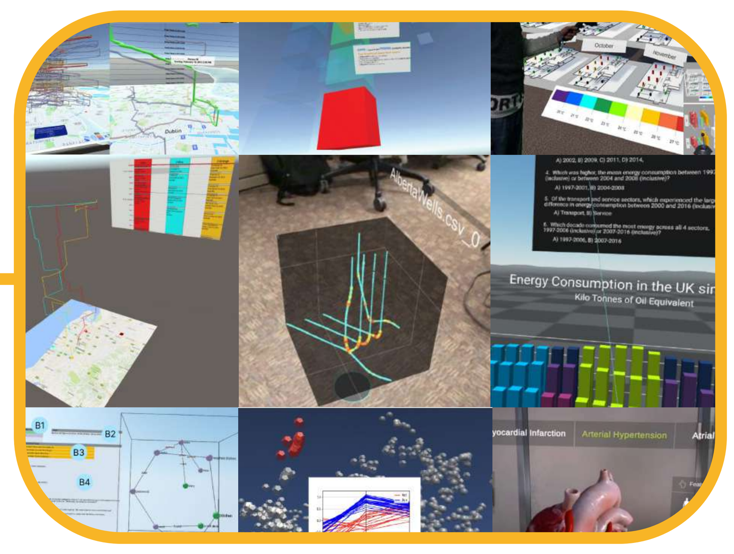
Research Introduction and Process

Since the growth of XR technologies, there has been a surge in the development of dashboards designed for immersive environ– ments. These dashboards handle complex data visualization within XR, requiring unique design considerations.



However, applying traditional dashboard de– signs directly to XR environments can lead to cognitive overload. The need for customized design approaches in XR dashboards has become evident to enhance user experience and interaction.

To address this, we collaborated with ex– perts to create a taxonomy that categorizes XR dashboard designs by factors like Type of Data, Audience Settings, Dashboard Im– plement, Purpose of Dashboard, Dashboard Design, Spatial Design.

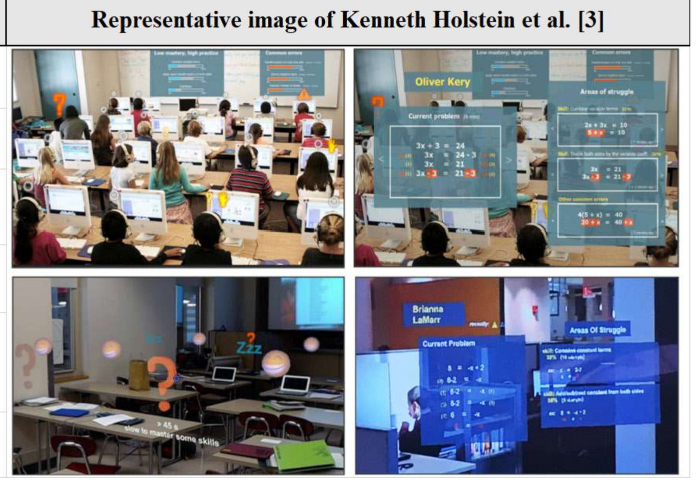


Using this taxonomy, 30 XR dashboard cases were analyzed to identify common design el– ements. The findings were then used to pro– pose an exploration system that allows users to navigate and apply these design insights in future XR dashboard projects.

Case Analysis Example

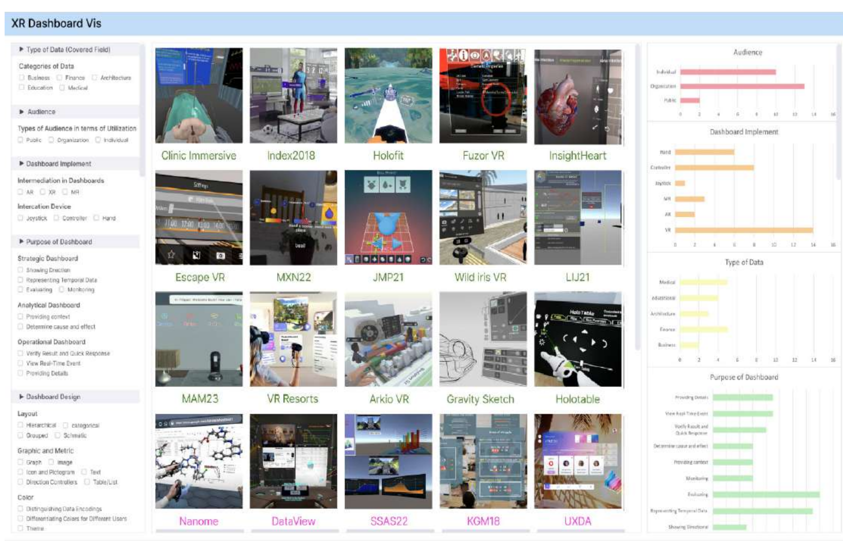
Main category	Classification result of Kenneth Holstein et al. [3]
Type of Data (Covered Field)	Categories of Data: **Educational**
Audience	Types of Audiences in terms of Utilization: **Organization, Individual**
Dashboard Implement	Intermediation in Dashboards: **AR** Interaction Device: **Hand**
Purpose of Dashboard	Strategic Dashboard: **Evaluating, Monitoring** Interaction Device: **Determine cause and effect**
Dashboard Design	Layout: **Categorical** Graphic & Metric: **Graph, Icon and Pictogram, Text** Color: **Distinguishing Data Encodings**
Spatial Design	Dashboard Layer: **Single Layer** Dashboard Placement: **Target Centered Lock**

Representative image of Kenneth Holstein et al. [3]

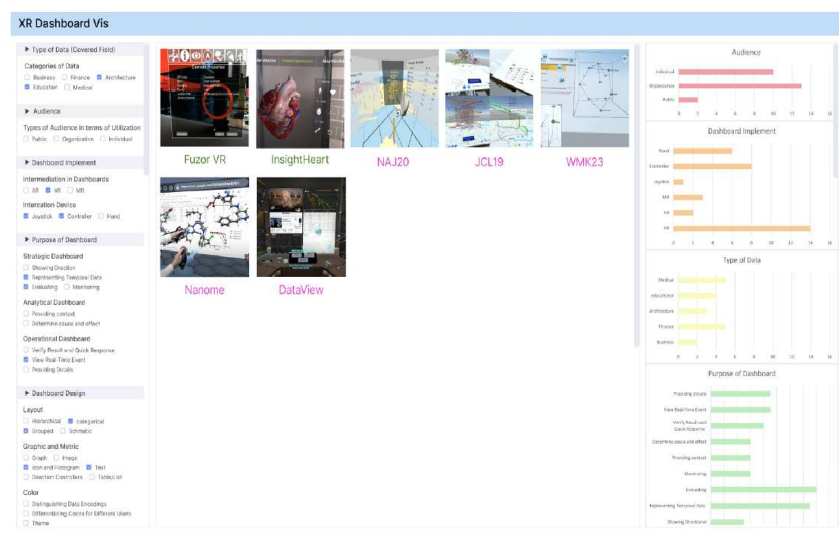


Classification result of Kenneth Holstein et al. (See reference 3 in the abstract paper)

Future Research (Exploration System)

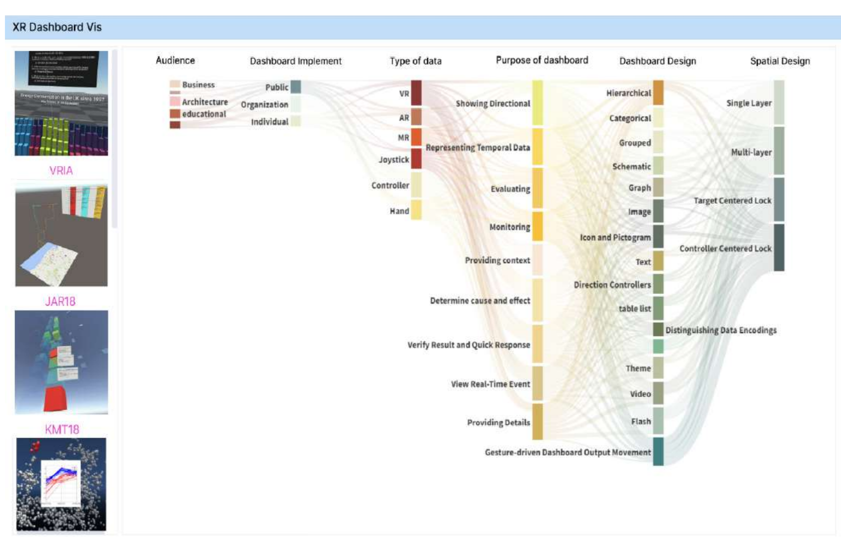


(a) Overview

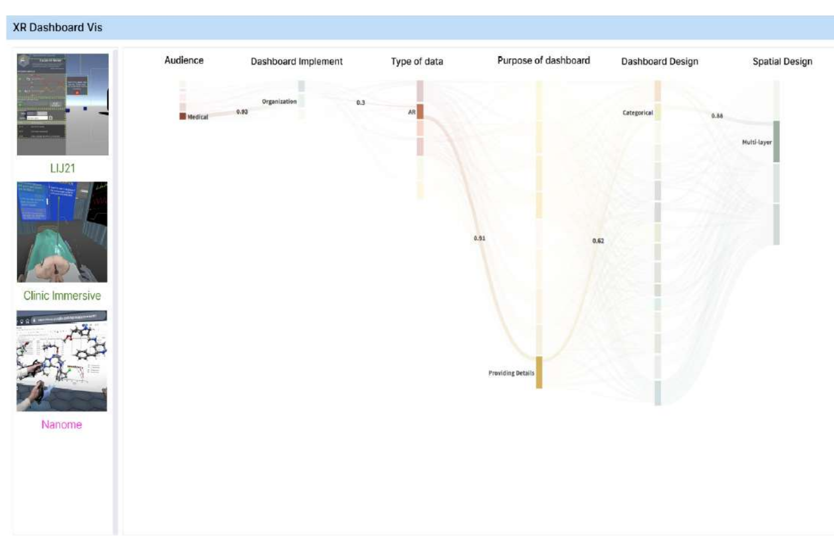


(b) Example of extracting dashboard cases based on a combination of taxonomy criteria

Exploration system concept design 1: Image-based faceted search mode



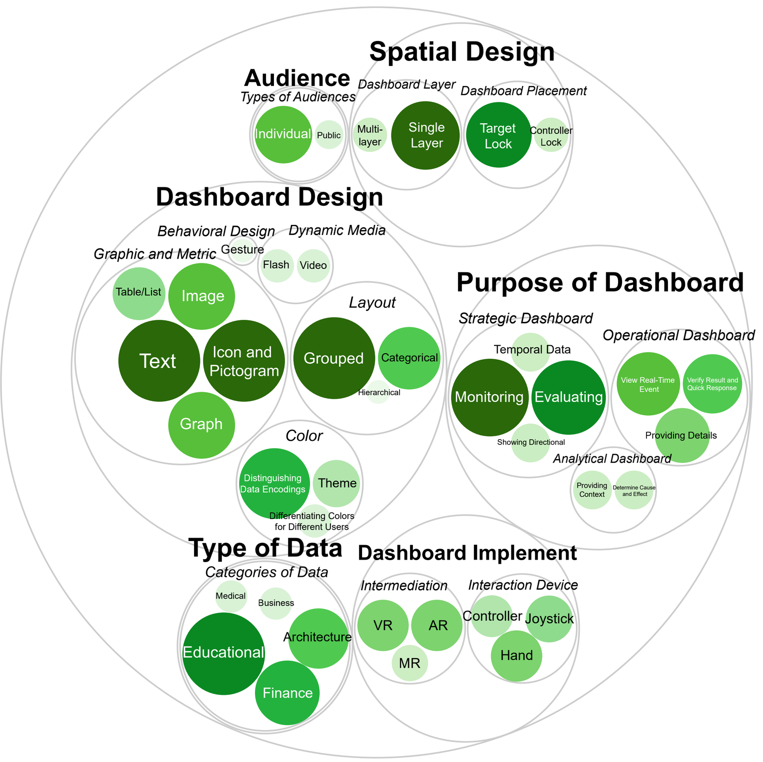
(a) Overview



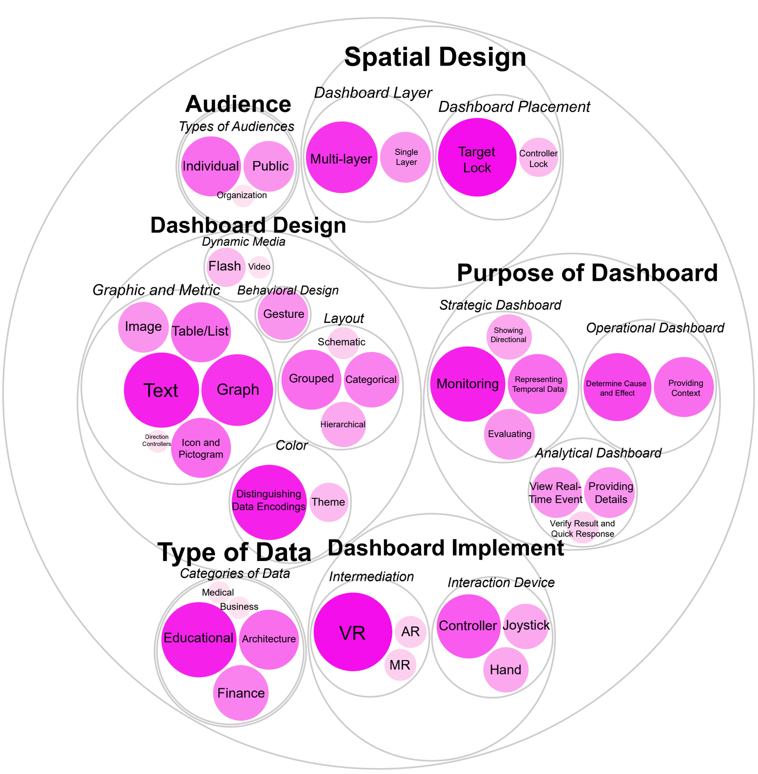
(b) Example of extracting dashboard cases based on a combination of taxonomy criteria

Exploration system concept design 2: Data visualization-based mode

Case Frequency Report



XR Content



XR-based DataVisTool